Mathematical Excalibur — Volume 7, No 3

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Example 1.

Numbers $1, 2, 3, \ldots, 1974$ are written on a board. You are allowed to replace any two of these numbers by one number, which is either the sum or the difference of these numbers. Show that after 1973 times performing this operation, the only number left on the board cannot be 0.

(1974 Kiev Math Olympiad)

Solution to Example 1.

Initially there are 987 odd numbers and 987 even numbers.

- If we choose to replace two numbers of the same parity, their sum and difference is even, so we either lose two odd numbers or we don't lose any odd numbers.
- If we choose to replace two numbers of different parity, their sum and difference is odd, so the number of odd numbers in the list doesn't change.

In each of these cases, the parity of the number of odd numbers on the board doesn't change and so it is an invariant. Thus the last number left on the board must be odd, and so it cannot be 0. ■

Example 2.

In an 8×8 board, there are 32 white pieces and 32 black pieces, one piece in each square. If a player can change all the white pieces to black and all the black pieces to white in any row or column in a single move, then is it possible that after finitely many moves, there will be exactly one black piece left on the board?

Solution to Example 2.

Let's say that in one row or column, there are w white pieces and b black pieces. Indeed $w + b = 8 \equiv 0 \pmod 2$. Thus $w \equiv b \pmod 2$. The total number of black pieces on the board will increase by $w - b$ each move, and similarly the total number of white pieces on the board will increase by $b - w$ each move. Since both $b - w$ and $w - b$ have to be even, the number of white pieces on the board at any given time is even, and the same for the number of black pieces on the board at any given time. Thus there can never only be 1 black piece left on the board. ■

Example 3.

Four x's and five o's are written around the circle in an arbitrary order. If two consecutive symbols are the same, then insert a new x between them. Otherwise insert a new o between them. Remove the old x's and o's. Keep on repeating this operation. Is it possible to get nine o's?

Solution to Example 3.

- If we have two consecutive x's, then we will replace them with a singular x, and the number of o's doesn't change.
- If we have two consecutive o's then we will replace them with a singular x, and the number of o's will decrease by 2.
- If we have an x next to an o then we will replace them with a singular o, and the number of o's doesn't change.

In each case the number of o's decreases and so it is impossible to get nine o's. ■

Example 4.

There are three piles of stones numbering 19, 8 and 9, respectively. You are allowed to choose two piles and transfer one stone from each of these two piles to the third piles. After several of these operations, is it possible that each of the three piles has 12 stones?

Solution to Example 4.

Modulo 3 the piles of stones are 1, 2 and 0 respectively. If a pile receives two stones, or if it has one stone removed this corresponds to adding 2 to the pile modulo 3. Since after each operation each of the piles modulo 3 change by the same amount, we can never have them all be equal to 0 mod 3 at some point, thus it is never possible to have each of the three piles with 12 stones. ■

Remark: The motivation for mod 3, comes from the fact that $2 \equiv -1 \ (mod\ 3)$.